

DAVID HYUNYOO JANG

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EDUCATION

Seoul National University

PharmD

Selected Coursework: Biochemistry · Pharmaceutical Statistics · Drug Delivery Systems

Mar 2022 – Feb 2028 (Expected)

Seoul, South Korea

Korea University

Coursework in Materials Science and Engineering (no degree)

Selected Coursework: Organic Nanomaterials · Fundamentals of Data Science

Mar 2019 – Aug 2021

Seoul, South Korea

RESEARCH EXPERIENCE

AI Protein Design Lab, Korea Advanced Institute of Science and Technology (KAIST)

Undergraduate Researcher (advisor: Gyu Rie Lee)

Jun 2025 – Aug 2025

Daejeon, South Korea

Collaborative small-molecule sensor binder design campaigns

open-source repos

- Developed, standardized, and released a reproducible end-to-end binder design workflow during phenytoin and carbamazepine sensor-design campaigns, integrating diffusion-based backbone generation, LigandMPNN, Rosetta scoring, AF3/Boltz-2 refolding, PLACER, TMalign, and RMSD-based triage with parquet-based metric lineage and SLURM-scalable execution. [\[code\]](#)
- Developed metric-guided LigandMPNN-FastRelax recycling to diversify and optimize candidate pools beyond sequence-only sampling across phenytoin/carbamazepine campaigns; in the carbamazepine campaign, expanded screened candidates (21→275) and improved the best Rosetta interface $\Delta\Delta G$ score ($-25.4 \rightarrow -31.6$). [\[code\]](#)

RFdiffusion2 atom-wise RASA conditioning for ligand orientation control

submitted upstream PR

- Implemented inference-time atom-wise RASA conditioning for RFdiffusion2 to control buried/exposed ligand atoms and improve ligand-orientation control while preserving compatibility with global RASA inputs. [\[PR\]](#)

Laboratory of Computational Drug Discovery, Seoul National University

Undergraduate Researcher (advisor: Juyong Lee)

Jul 2023 – Aug 2025

Seoul, South Korea

PREpiBind — pMHC-II binding: representation benchmark & web server

first-author; manuscript in prep

- Independently built PREpiBind, an open framework benchmarking 10 protein representations — BLOSUM62, structure-prediction embeddings extracted by modifying AlphaFold3/Chai-1/Boltz-1 inference code, and protein language models (ESM C/ESM3) — for pMHC class II binding on 205K IEDB entries; PLM embeddings consistently outperformed structure and alignment representations (counter to the structure-first hypothesis) and matched or exceeded NetMHCIIpan-4.3, MixMHC2pred-2, and DeepNeo.
- Established leakage-controlled splits and stress-tested generalization under leave-one-molecule-out and cross-species (mouse) validation, where PLM embeddings generalized best among the benchmarked representations; re-implemented DeepNeo from scratch in PyTorch (matching its reported accuracy) for a fair head-to-head.
- Selected ESM C 300M for the best accuracy–efficiency trade-off and released open tools — pre-trained models, Colab/CLI demos, and a public [web server](#) covering 7,282 IPD-IMGT/HLA alleles zero-shot plus a custom-sequence mode for non-human alleles — with inference optimized to up to $\sim 69\times$ faster than command-line tools. [\[code\]](#)

LLMs on the Korean Pharmacist Licensing Exam

under review

- Designed and built an end-to-end benchmark of 27 LLMs — 7 proprietary and 20 open-source (4–104B) — across six KPLE exam years (2019–2024) in Korean and English, covering question extraction (text + vision-OCR), LLM-based translation, local/API inference, and official scoring with human-percentile normalization.
- 7 models passed all years in both languages — the top reaching the top 12% of human examinees — under a logarithmic size–performance trend; developed a multi-pronged data-contamination analysis arguing against pretraining leakage as the dominant performance driver. [\[code\]](#)

CB1 receptor small-molecule virtual screening

- Reproduced an end-to-end structure-based virtual screen of the Enamine Hit Locator Library (460K compounds) against the CB1 receptor on an 8-GPU cluster (~ 12 h) — self-compiling AutoDock-GPU from source for multi-GPU execution (the prebuilt binary used 1 of 8 GPUs) — and automating batch docking, result parsing, and ranking.

Structure & Knowledge Injection into Machine Learning Lab, Seoul National University

Undergraduate Researcher (advisor: Jay-Yoon Lee)

May 2024 – Sep 2024

Seoul, South Korea

Semi-supervised data augmentation for RoseTTAFold2NA collaboration with Minkyung Baek's lab

- Implemented the training pipeline for a semi-supervised joint-loss objective that augments RF2NA with unlabeled self-distilled structures — built the mixed labeled/unlabeled data loader, per-batch loss routing, and a distributed sampler with tunable unlabeled fraction, consolidating scattered prototype scripts into one parameterized training entry point.

- Designed and ran a controlled low-label study — PDB-only vs distilled-augmentation vs joint-loss semi-supervision across 10/30/50% label-density regimes — to test whether unlabeled data improves RF2NA learning under sparse supervision.

Systems Pharmacology Lab, Seoul National University
Undergraduate Internship (advisor: Minsoo Noh)

Jun 2023 – Jul 2023
Seoul, South Korea

- Performed HaCaT cell culture and PGE₂ competitive ELISA to assess anti-inflammatory effects of tryptophol; conducted dose–response analysis and benchmarking against NSAIDs.

PAPERS AND PRESENTATIONS

- [1] PREpiBind: Protein Representation-integrated Epitope-MHC Class II Binding Prediction

David Hyunwoo Jang, Dongwoo Kim, Byungho Park, Untaek Hwang, Yoonjoo Choi, Juyong Lee
 KIDDS 2025 Summer Workshop (Korean In silico bioDesign and Discovery Society, Poster)
Manuscript in preparation
- [2] Proprietary and Open-Source Large Language Models on the Korean Pharmacist Licensing Examination: A Comparative Benchmarking Study

David Hyunwoo Jang, Juyong Lee
 Under review
- [3] DeepXeno: Structure-Informed Prediction of Xenotransplantation Immunogenicity

David Hyunwoo Jang, Byungho Park, Untaek Hwang, Juyong Lee
 Outstanding Research Presentation in 2024 SNU Student-Directed Education Undergraduate Research Program (Talk)

AWARDS & FUNDING

Research Excellence Award, 2024 SNU Student-Directed Education Undergraduate Research Program
President of Seoul National University

2025
 2024 SNU Student-Directed Education Undergraduate Research Program
Faculty of Liberal Education, Seoul National University

- Awarded to two projects per fields (ten total) for excellence in research execution and outcomes.
- Selected as one of 36 projects to receive funding for undergraduate research. (*KRW 7M*)

LEADERSHIP & COMMUNICATION

SNU Alpha | Deep Learning Study Club
Executive Member, Co-Founder

Spring 2022 – Winter 2024
College of Pharmacy, Seoul National University

- Peer-taught deep learning fundamentals and applications in computational drug discovery to pharmacy students.
- Led weekly study sessions on Natural Language Processing (Fall 2023) and PyTorch (Spring 2024).

Military Service
Squad Leader, Sergeant

Aug 2025 – Feb 2027 (Expected)
III Corps, Republic of Korea Army

- Led a 10-person squad at an air-defense emplacement, coordinating daily operations and training.
- Operated and maintained air-defense radar systems in a frontline sector along the Korean DMZ.

Seoul National University Hospital
Hospital Pharmacy Practicum

Dec 2024 – Feb 2025
Seoul, South Korea

- Completed rotations in 12 specialty departments, gaining broad exposure to hospital pharmacy practice.
- Conducted drug therapy case reviews and provided patient counseling in multidisciplinary clinical settings.

E-Blue Pharmacy
Community Pharmacy Practicum

Feb 2025 – Mar 2025
Suwon, South Korea

- Screened prescriptions for errors and performed medication therapy management interventions.
- Conducted patient counseling for prescription medications and over-the-counter consultations.

TECHNICAL SKILLS

Programming & Data: Python, PyTorch, Numpy, Bash, pandas, matplotlib, Streamlit, Nginx
Modeling & Docking: RDKit, PyMOL, Chimera, AutoDock, PyRosetta, RFdiffusion (AA), LigandMPNN, AlphaFold3
Systems & DevOps: Linux, Git, Slurm, conda/mamba, HuggingFace
Experimental Techniques: cell culture (HaCaT), buffer/reagent prep, competitive ELISA, serial dilutions, plate reader